INTERNAL PRE-READ

RETICLE Co-essentiality Network

A gene–gene functional network built **purely from BioGRID CRISPR screens** — no literature, no GO, no STRING. This version spells out the actual math at every step and defines every acronym on first use, so it should be readable with a biology background alone, no prior exposure to this pipeline needed.

Two genes are connected when knocking each one out perturbs cells the **same way** across BioGRID's ~1,700 published CRISPR screens, pooled together. Every number below is computed by a short, **fully deterministic** pipeline (fixed-seed only, no LLM at compute time) straight from one table of screen measurements. Validated by recovery of independently-known protein complexes (CORUM), which it does **better** than any single screen-type alone.

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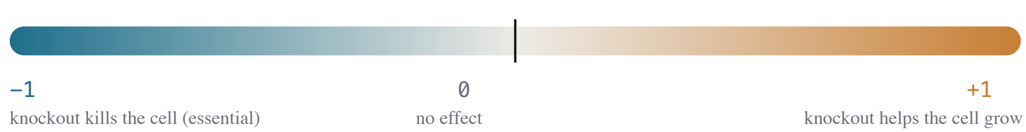
1 · The data — one harmonized table

The only upstream data source is BioGRID ORCS, a public archive of published CRISPR screen results. Different labs score their screens with different statistics — `MaGeCK`, `CasTLE`, `Log2FC`, `STARS` — whose raw numbers aren't on the same scale and can't be compared directly. RETICLE's harmonization layer (`harmonize_scores.py`, upstream of this document) fixes that by converting every score to a **rank within its own screen**:

$$\mathrm{PERCENTILE\_SCORE}(g, s) = \frac{2\ \mathrm{rank}(g, s)}{\mathrm{max\_rank}(s)} - 1$$

where `rank(g,s)` is gene *g*'s rank (average rank if tied) among all genes with a valid score in screen *s*, 0-indexed. This maps every screen onto the same **[–1, +1]** scale regardless of the original statistic — only relative position within a screen matters, never the raw magnitude.



PERCENTILE_SCORE: a gene's rank position within one screen, mapped to [–1,+1]

hit / `IS_HIT` — a 0/1 flag the original screen's authors assigned: 1 means *they* judged this gene's effect to be statistically significant in their own analysis (their own p-value/FDR cutoff), not something RETICLE computes. Across the whole dataset, hits are rare: **6.7%** of all (gene, screen) pairs. The other 93% is the "noisy middle" — small, inconsistent effects.

2 · A gene's "fingerprint"

Lay the data out as a grid — rows = genes, columns = screens, each cell = that gene's `PERCENTILE_SCORE` in that screen. One gene's **entire row** — its scores across every screen it was measured in — is its **fingerprint**: a numeric summary of how that gene behaves under many different perturbations.

3 · The core rule — connection = fingerprints that co-vary

Two genes are connected when their fingerprints **rise and fall together**. The tool for measuring "rise and fall together" is the Pearson correlation coefficient, `r` — a standard statistic, not something specific to this project:

$$r(i,j) \;=\; \frac{\sum_s (x_{i,s} - \bar{x}_i)\,(x_{j,s} - \bar{x}_j)}{\sqrt{\sum_s (x_{i,s} - \bar{x}_i)^2 \;\cdot\; \sum_s (x_{j,s} - \bar{x}_j)^2}}$$

$x_{i,s}$ = gene *i*'s PERCENTILE_SCORE in screen *s*; $\bar{x}_i$ = gene *i*'s own average across the screens being compared. `r` ranges –1 (perfectly opposite) to +1 (perfectly together); 0 = no relationship.
Implementation note: a gene isn't measured in every screen, so missing entries are filled with that gene's own mean $\bar{x}_i$ before the sums are taken — a value equal to the mean contributes exactly zero to the $(x - \bar{x})$ terms, so an unmeasured screen simply drops out of the comparison rather than pulling the correlation either way.

Subtracting each gene's own average (**centering**) is what makes this work: the question becomes *"when gene A is more essential than ITS OWN typical level, is gene B also more essential than ITS OWN typical level?"* — not "are both genes essential." That distinction matters:

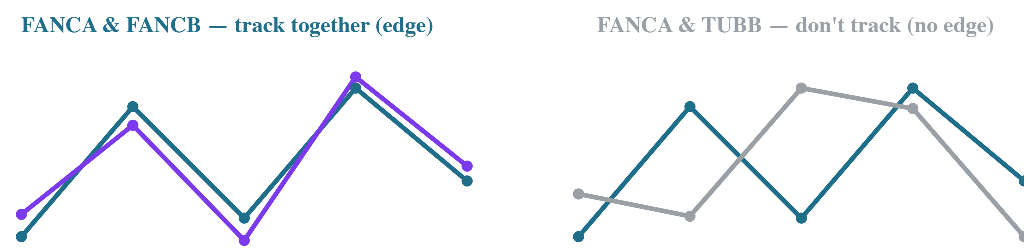
WORKED EXAMPLE

Percentile scores across 5 screens (more negative = more essential in that screen):

gene	screen 1	screen 2	screen 3	screen 4	screen 5
FANCA	−0.9	−0.3	−0.8	−0.2	−0.7
FANCB	−0.8	−0.4	−0.9	−0.1	−0.6
TUBB	−0.7	−0.8	−0.2	−0.3	−0.9

FANCA and FANCB dip and rise **on the same screens** (1,3,5 down · 2,4 up). TUBB is *also* essential on average (cells need their cytoskeleton), but it dips and rises on **different screens** — a real gene, just not correlated with FANCA here.

`r(FANCA, FANCB) = +0.94 → edge kept`   `r(FANCA, TUBB) = +0.10 → no edge`



the correlation compares shape (which screens spike), not overall level

4 · Pooling every screen (and what that trades off)

We compute one correlation per gene pair over **every** FULL screen at once — all ~1,700 human screens, spanning fitness, stress, drug-response and reporter assays. This is the standard co-essentiality approach (e.g. DepMap): more screens = more statistical power, and constitutive complexes (ribosome, proteasome, spliceosome…) surface regardless of what any single screen happened to be measuring. On the CORUM benchmark (§9) the pooled network scores **higher** than any single screen-type computed on its own.

The honest tradeoff. Pooling answers *"are these genes constitutive partners?"* It deliberately **averages away** relationships that exist only under one condition — on a hand-checked set of pairs that are strong under DNA-damage yet flat in plain fitness, computing edges within a single context recovered roughly **2× more** of them than the pool. So this version is the **constitutive backbone**; a context-resolved layer (edges recomputed within one assay type or drug mechanism) is a deliberate future add-on, noted in §*Status*.

5 · Pitfall #1 — which genes even qualify as nodes

Problem. Only ~6.7% of (gene, screen) rows are hits. A gene *never* called a hit has a flat, noisy fingerprint — no evidence it does anything. And a gene measured in only a handful of screens gives an unstable correlation (its normalization rests on too few points). Both can spuriously correlate with anything.

Fix — two floors. A gene is a node only if it was a hit in ≥ 5 screens **and** was measured in ≥ 100 screens:

$$\sum_s \mathrm{IS\_HIT}(g,s) \;\geq\; 5 \qquad \text{and} \qquad \#\{\, s : g \text{ measured in } s \,\} \;\geq\; 100$$

The hit floor drops genes with no phenotype anywhere (mostly lincRNA / uncharacterized); the measurement floor drops sparsely-measured genes whose correlation would be unstable. Together they leave ~19,000 well-evidenced genes. (Pooling helps here too: the median node is now measured in ~1,380 of the ~1,700 screens, so almost every gene clears the floor comfortably.)

6 · Pitfall #2 — "hub" genes that correlate with everyone

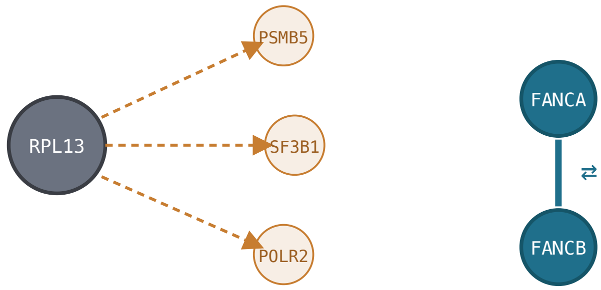
Problem. Pan-essential genes (ribosome, proteasome subunits) drop out in nearly every proliferation screen, so they end up moderately correlated with a large number of unrelated essential genes — smearing the graph into one big blob.

Fix — reciprocal-rank. For gene *i*, define its **top-25** as the 25 partners with the highest `r(i,j)`, subject to two floors:

$$\begin{aligned} \mathrm{top25}(i) \;&=\; \{\, j \;:\; r(i,j) \text{ among the 25 largest},\;\; r(i,j) \geq 0.15,\;\; \mathrm{support}(i,j) \geq 20 \,\} \\ \mathrm{reciprocal}(i,j) \;&=\; 1 \quad \iff \quad j \in \mathrm{top25}(i) \;\wedge\; i \in \mathrm{top25}(j) \end{aligned}$$

$\mathrm{support}(i,j)$ = number of screens where *both* genes have a PERCENTILE_SCORE; it guards against a high correlation computed over too few shared screens (noise). An edge with only $j \in \mathrm{top25}(i)$ but not the reverse is kept in the data, just flagged non-reciprocal.

Why this works: a hub points to many genes, but *those genes'* own top-25 is dominated by their real, specific complex-mates — the hub rarely makes it back into their list. A true functional pair (FANCA↔FANCB) is each other's best match in both directions and survives. This uses no external annotation — only the ranking that falls out of the data itself.



only mutually-preferred edges survive as the default "reciprocal" view

7 · The noise threshold, and what gets stored

Problem. Pure noise still produces a correlation, of typical size $|r| \approx 1/\sqrt{\#\text{screens}}$. Picking a cutoff by eye is arbitrary — set it too low and the edge list fills with coincidences (this bites hardest when few screens are pooled).

Fix — a shuffle null. We build a null by shuffling values *among each screen's measured genes* (destroys cross-screen co-movement, preserves every gene's coverage/support and each screen's distribution — the correlation you'd see from noise alone), then raise the threshold to the lowest r at which the reciprocal-edge FDR — (null edges above the cut) / (real edges above the cut) — drops to ≤ 5%. Over the full ~1,700-screen pool the noise floor is tiny (≈ 0.024), so the threshold lands just above the 0.15 floor, at **r ≥ 0.16**. The shuffle uses a fixed seed → reproducible. (The fewer screens you pool, the more this matters — it's what would keep a future per-context layer honest.)

Restricting to genome-wide (`FULL`) screens — where every gene got a score, unlike `HIT_ONLY` screens that only report hits and can't support a correlation — each surviving edge is written to `net_edge` with columns `(gene_a, gene_b, r, support, reciprocal)`.

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8 · Design choices tested and rejected

Every "smarter" alternative below was measured against the CORUM benchmark (§9) before this method was chosen — the simple version won on the numbers, not by default.

✗ **Hit-anchoring** (correlate only over screens where at least one gene is a hit) — AUROC dropped 0.07–0.10. With hits this sparse, the "anchor" screens are mostly one-gene-hit / other-gene-mid, which dilutes rather than sharpens the signal; core-essential pairs also lose variance once restricted to their most extreme screens.

✗ **CLR / top-PC removal** (the standard DepMap-style hub correction — normalize each gene's correlations against a background distribution) — hurts AUROC genome-wide: it inflates correlations to low-background "loner" genes (uncharacterized genes with almost no true partners) and demotes real complex-mates. Reciprocal-rank on the raw correlation outperforms it here.

9 · Validation

CORUM — a manually-curated public database of experimentally-verified human protein complexes (i.e. which proteins physically assemble together to do a job). It is **independent** of this pipeline and not derived from literature co-occurrence the way STRING is, so it's a clean ground truth to test against.

Question: do gene pairs from the *same* CORUM complex get a higher correlation than pairs from *different* complexes? Measured with AUROC (Area Under the ROC Curve):

$$\text{AUROC} = P\big(r(\text{same-complex pair}) > r(\text{different-complex pair})\big)$$

0.5 = the two groups are indistinguishable (chance); 1.0 = every same-complex pair outranks every different-complex pair.

0.717	2,626	7.2M	18,999
CORUM AUROC (pooled)	CORUM complexes tested (3–60 subunits)	gene pairs scored	genes in the network

Comfortably above the 0.5 chance line, and *higher* than any single screen-type computed alone (fitness-only 0.684, DNA-damage-only 0.661) — pooling wins for constitutive complexes. Honestly a floor, too: CORUM documents only a fraction of all real functional relationships, so some "false positives" here are likely true, just uncatalogued.

Clean recovery of known complexes

Each hub gene's strongest reciprocal partners land squarely in its real complex:

RPL13 (ribosome)	RPL3, RPL7, RPL11, RPL4, RPL34, IMP3 — all ribosomal
PSMB5 (proteasome)	PSMB4, PSMA5, UFD1 — proteasome / ubiquitin
MED1 (Mediator)	MED9, MED7
FANCD2 (Fanconi anemia)	FANCA, FANCB, FANCC, FANCE, FANCL, UBE2T — the Fanconi core

Pooling ~1,700 *diverse* screens gives enough resolving power to **separate** ribosome from proteasome — two machines that co-drop together in any single fitness screen and blur into one blob there.

10 · Design principles

Pure CRISPR. No PubMed, no GO (Gene Ontology — a standard functional-annotation database), no STRING (a widely-used protein network built substantially from literature co-occurrence and database annotation) enters edge computation. This is the deliberate point of differentiation from literature-derived networks, and it's also why CORUM (biochemistry, not literature attention) is the right validation set.

Constitutive by design. Edges are pooled across all ~1,700 screens — this surfaces genes that partner *constitutively* and gives the most statistical power. Relationships specific to a single condition are intentionally averaged out here; recovering them is the job of the planned context-resolved add-on layer.

Observations are truth; edges are derived. Nothing is stored that can't be recomputed from the raw `harmonized_scores` table with full provenance back to specific screens.

Deterministic and auditable. No LLM anywhere in edge computation; the only randomness is a **fixed-seed** permutation used to set the noise threshold — so identical input still produces byte-identical output, and every edge traces to specific `(screen, gene, percentile)` rows.

Status & what's next

Network	`all` (pooled): 1,745 FULL human screens · 18,999 nodes · 445,617 edges (17,337 reciprocal), threshold r≥0.16
Interface	interactive web viewer — search a gene, reciprocal-only filter, click to recenter
Scope	human only (v1)

Next: add a co-hit channel (Fisher's exact test) for `HIT_ONLY` screens, which have no denominator for a correlation; and a **context-resolved layer** — edges recomputed within one assay type or drug mechanism — to recover the condition-specific relationships the pool averages out (§4).

Glossary — every non-obvious term used above

`BioGRID / ORCS`	Public archive of published CRISPR screen datasets — the sole upstream data source.
`Screen`	One CRISPR knockout experiment: every gene knocked out (or a large subset), effect on cells measured.
`MaGeCK / CasTLE / Log2FC / STARS`	Different statistical methods labs use to score a screen; not comparable across labs on raw scale, hence the rank-based PERCENTILE_SCORE.

Hit / `IS_HIT`	The original screen authors' own call that a gene's effect was significant — inherited from the source data, not computed by me.
`FULL` vs `HIT_ONLY` screen	`FULL` = every gene got a score (needed for correlation); `HIT_ONLY` = only significant hits were reported, no non-hit baseline.
Pearson correlation (r)	Standard statistic for how closely two numeric sequences move together, from −1 to +1.
Support	Number of screens where both genes in a pair were actually measured together.
Reciprocal-rank	An edge kept only when each gene is among the other's top correlated partners — the hub-control mechanism.
CORUM	Curated database of experimentally-verified protein complexes, used only for validation, never for building edges.
AUROC	Area Under the ROC Curve — probability a random true pair scores higher than a random false pair; 0.5 = chance, 1.0 = perfect.
GO (Gene Ontology)	A standard controlled vocabulary / database of gene functions, curated substantially from literature.
STRING	A widely-used gene/protein network built from literature co-occurrence, databases, and experiments combined — the "literature-based" comparison point for this project.
FDR	False Discovery Rate — a statistical correction applied when testing many hypotheses (e.g. many gene pairs) at once, to control spurious findings.
